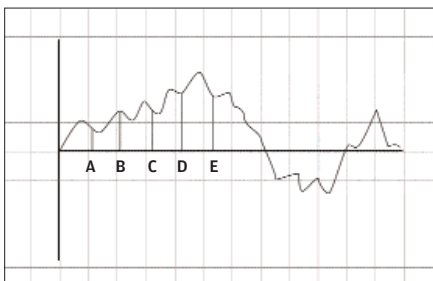


A high sampling rate means we look at it more times per second, at points A, B, C, D, and E, as below:

The bitrate is the number of bits used to hold the data captured every second. Music is sampled at rates like 44.1 kHz, with 16 bits or 24 bits (or so) used to hold the data *at each sample*. So the bitrate of music sampled this way would be



$44,100 \times 16 \times 2 = 1,411,200$ bits per second, or 1,378 kbps, or 1.35Mbps.

Some typical sampling rates:

44.1 kHz CD, DAT

48 kHz DVD-Video, DV camera, DAT

96 kHz DVD-Audio

2.2 Your MP3s: Good Enough?

Consider this: cymbals (our reference point for high frequencies!) can be as high as 10 kHz (our ears can hear sounds between 20 Hz and 20 kHz). Now, think of the sampling as the frames in a movie, which, as you know, are usually taken 24 times a second. That is, there are 24 frames that you see every second. If we're sampling the cymbals at 44.1 kHz, only approximately four "frames" per second are representing the cymbals. In contrast, the human voice is somewhere around 1 kHz, and that's 40 "frames" per second. Hence, high frequencies need a higher sampling rate to be represented more accurately in the digital format.

Now, what we mentioned above—44.1 kHz sampling at 16 bits—comes to 1378 kbps. That's something like CD quality (though strict-